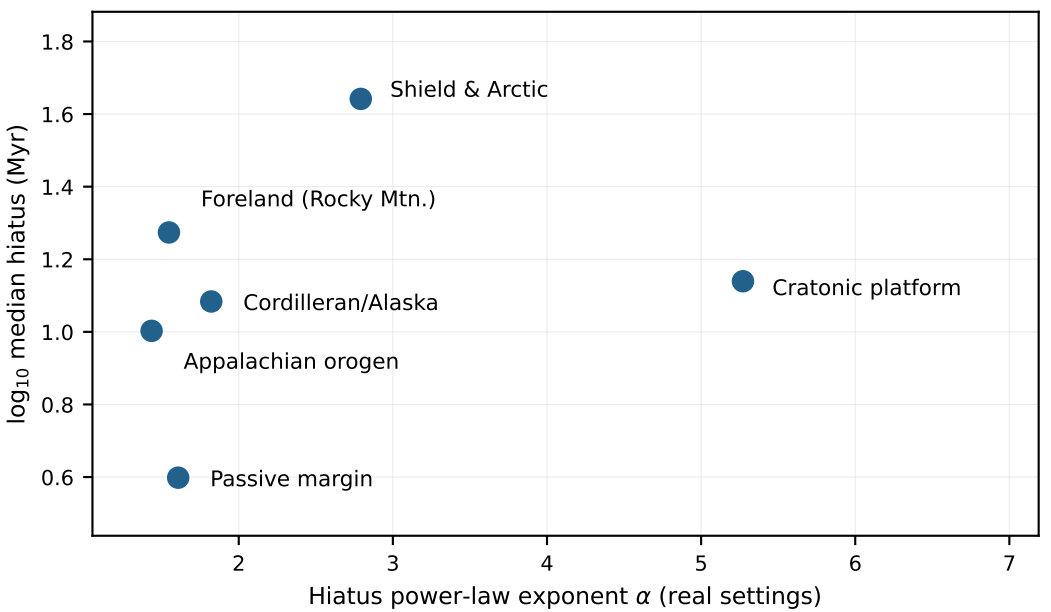
(a) Mechanism × diagnostic evidence matrix

	Heavy-tailed hiatus CCDF	Low completeness at fine τ	Low preservation efficiency	Power law beats exponential (LLR)
F1 white-noise forcing	yes	yes	yes	yes
F2 multi-periodic forcing	yes	partly	partly	yes
F3 long-memory forcing	yes	no	no	yes
Real: orogenic/foreland/passive	yes	partly	partly	yes
Real: cratonic platform	partly	yes	partly	no
Real: shield & arctic	yes	partly	partly	partly

Compiled from Figs. 4, 5, 7, 8; 'partly' = intermediate or range-limited behaviour.

(b) Setting-level hiatus statistics



(c) Prioritised research roadmap

1. Global hiatus databases with uncertainty-aware ages (extend beyond N. America)
2. Age-depth models that explicitly allow hiatuses (beyond smooth priors)
3. Coupled landscape-stratigraphy models with mass tracking and recycling
4. Independent constraints on eroded mass (thermochronology, provenance)
5. Community protocols for power-law testing of geologic durations
6. Assimilation of real forcing records into inverse stratigraphic models